

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, IV: THE LIMIT MODULE OF ABELIAN TOWERS AND THE CHARACTERISTIC IDENTITY

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ABSTRACT. For an abelian $\mathbb{Z}/\ell^k$ voltage tower (Y_k) over a finite base graph, Papers II–III established the exact Iwasawa growth law of the sandpile orders and its Newton-polygon mechanism, leaving the limit-module statement of [II, Conj. 3.6(ii-a)] open. This paper closes it. Writing $D(t)$ for the voltage Laplacian pencil and $\Lambda = \mathbb{Z}_\ell[[T]]$, we prove:

$$X_\infty := \varprojlim_k (\operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell, \pi_*) \cong \Lambda^n / D(1+T) \Lambda^n,$$

a finitely generated torsion Λ -module with $\operatorname{char}_\Lambda(X_\infty) = (\det D(1+T))$, together with an *exact* control theorem $X_\infty / \omega_k X_\infty \cong \operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell$ — exact, not merely pseudo-isomorphic, because the tower’s presentations are literally the ω_k -specializations of one Λ -presentation. The proof rests on three structural identities — the derived Laplacian is the block-circulant $D(\sigma)$; the covering pushforward is the group-ring coefficient projection; pushforward is surjective on ℓ -primary torsion with kernel of the exact predicted order — each proved and additionally machine-verified at every level of two towers. Combined with [II, Rem. 5.4], the measured invariants $(\mu, \lambda) = (\mu(g), \lambda(g) - 1)$, $g = \det D(1+T)$, are now consequences of a theorem rather than observations. The result is stated with explicit deference to the graph-Iwasawa literature of Vallières, McGown–Vallières and Gonet, whose verified titles indicate substantial overlap; what is new here, at minimum, is the exactness of the control isomorphism at the level of Smith normal forms and the integration into the trilogy’s K -theoretic framework. The non-normal congruence tower ([II, Conj. 3.6(ii-b)]) and the canonicity problem ([III, Rem. 1.10]) are untouched and remain the open frontier.

STATUS OF THE RESULTS

Every numbered statement in this paper is proved in full. Lemmas 1.1, 1.2 and 1.3 are elementary and were additionally verified by exact machine computation at every level $k \leq 3$ of the tower with voltages $(1, 1, 1)$, $\ell = 2$, and every level $k \leq 2$ of the tower $(1, 0, 0)$, $\ell = 3$ (Table 1). The single external algebraic input is the structure theory of finitely generated Λ -modules, quoted with a flagged citation. The relation of Theorem 2.1 to the published graph-Iwasawa literature is discussed, deliberately and prominently, in Section 4.

1. SETUP AND THE THREE STRUCTURAL IDENTITIES

Fix a finite connected base graph Y_0 on n vertices, a prime ℓ , a spanning tree, and a voltage assignment α on the chords, giving the derived $\mathbb{Z}/\ell^k$ -covers Y_k on $n\ell^k$ vertices with deck generator σ and covering maps $\pi: Y_{k+1} \rightarrow Y_k$. Let

$$D(t) = (q+1)I_n - A(t) \in M_n(\mathbb{Z}[t, t^{-1}])$$

Date: 12 September 2026.

2020 Mathematics Subject Classification. Primary 11R23, 05C50; Secondary 05C25, 13C12, 46L80.

Key words and phrases. Iwasawa module, voltage graph, sandpile group, characteristic ideal, control theorem, Bruhat–Tits tree.

Paper IV of the series; Papers I–III are cited as [I], [II], [III]. This paper upgrades [II, Conj. 3.6(ii-a)] from a verified growth law to a limit-module theorem. All computational claims are reproduced by the script `limit_module.py` in the ancillary bundle.

be the voltage Laplacian pencil ($A(t)$ the voltage adjacency), Δ_k the ordinary Laplacian of Y_k , $G_k = \mathbb{Z}/\ell^k$, and order V_{Y_k} as $(v, g) \mapsto vN + g$, $N = \ell^k$. Write S_N for the cyclic shift on $\mathbb{Z}_\ell^N$ and $Q_N: \mathbb{Z}_\ell^{\ell N} \rightarrow \mathbb{Z}_\ell^N$ for reduction of the group coordinate, $e_g \mapsto e_{g \bmod N}$.

Lemma 1.1 (Block-circulant identification). *In the ordering above, $\Delta_k = D(S_N)$ exactly: the level- k Laplacian is the image of the pencil under $t \mapsto S_N$. Equivalently, as a $\mathbb{Z}[G_k]$ -module with the deck action, $\text{coker}_{\mathbb{Z}}(\Delta_k) \cong \mathbb{Z}[G_k]^n / D(\sigma)\mathbb{Z}[G_k]^n$.*

Proof. This is the definition of the derived graph read backwards: a base edge (u, v) of voltage a contributes the arcs $(u, g) - (v, g+a)$ for all g , i.e. the block $E_{uv} \otimes S_N^a$ plus its transpose, and the degree term is $(q+1)I_n \otimes I_N$. Summing gives $\Delta_k = D(S_N)$; identifying $\mathbb{Z}_\ell^N \cong \mathbb{Z}_\ell[G_k]$ with $S_N =$ multiplication by σ gives the module form. Verified as an exact matrix identity at all computed levels. $\square$

Lemma 1.2 (Pushforward is coefficient projection). *Let $P_k := I_n \otimes Q_{\ell^k}$, the matrix of the covering pushforward $\pi_*: \mathbb{Z}^{V_{Y_{k+1}}} \rightarrow \mathbb{Z}^{V_{Y_k}}$ (sum over fibres). Then*

$$P_k \Delta_{k+1} = \Delta_k P_k, \quad P_k R_{k+1} = R_k P_k,$$

where $R_k = I_n \otimes S_{\ell^k}$ is the deck action; under Lemma 1.1 the induced map on cokernels is the group-ring coefficient projection $\mathbb{Z}[G_{k+1}] \rightarrow \mathbb{Z}[G_k]$. Moreover P_k is surjective.

Proof. Both identities reduce to the one-line facts $Q S_{\ell N} = S_N Q$ and $Q S_{\ell N}^a = S_N^a Q$ applied blockwise to $\Delta = D(S)$; surjectivity is immediate since every $e_{(v,h)}$ is hit. Verified as exact matrix identities at all computed levels. $\square$

Lemma 1.3 (Norm surjectivity on torsion). *The induced norm $\pi_*: \text{coker}_{\mathbb{Z}}(\Delta_{k+1}) \rightarrow \text{coker}_{\mathbb{Z}}(\Delta_k)$ is surjective; it preserves the degree functional $\deg$ (total mass), hence restricts to a surjection of torsion subgroups, with kernel of order*

$$|\ker(\pi_*|_{\text{Tor}_\ell})| = \ell^{\text{ord}_\ell |\text{Jac}(Y_{k+1})| - \text{ord}_\ell |\text{Jac}(Y_k)|}.$$

Proof. Surjectivity of P_k plus the intertwining gives surjectivity on cokernels. The column sums of Δ_k vanish, so $\deg([y]) = \sum_i y_i$ is well defined and realizes $\text{coker}_{\mathbb{Z}}(\Delta_k) \cong \mathbb{Z} \oplus \text{Tor}$ with $\text{Tor} = \ker(\deg)$ (a finitely generated group surjecting onto $\mathbb{Z}$ with rank-zero kernel has kernel equal to its torsion). Pushforward preserves total mass, so $\deg \circ \pi_* = \deg$; hence for torsion t downstairs any preimage a has $\deg(a) = 0$, i.e. a is torsion: surjectivity on Tor . The kernel order is then forced by counting. Verified, including the exact kernel orders ℓ^3, ℓ^3, ℓ^3 and ℓ^1, ℓ^1 at the computed transitions (Table 1). $\square$

2. THE LIMIT MODULE AND THE CHARACTERISTIC IDENTITY

Theorem 2.1 (Limit module). *Let $g(T) := \det D(1+T) \in \mathbb{Z}_\ell[[T]]$ and assume $g \neq 0$ (true in every tower of [II, §5]). Then, with transition maps the norms π_* of Lemma 1.3,*

$$X_\infty := \varprojlim_k (\text{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell) \cong \Lambda^n / D(1+T) \Lambda^n, \quad \Lambda = \mathbb{Z}_\ell[[T]],$$

under the standard identification $\varprojlim_k \mathbb{Z}_\ell[G_k] \cong \Lambda$, $\sigma \mapsto 1+T$ (the Laurent entries $t^{\pm a}$ of the pencil landing in $\Lambda^\times$ -multiples since $1+T \in \Lambda^\times$). The module X_∞ is finitely generated and torsion over Λ , and

$$\text{char}_\Lambda(X_\infty) = (g(T)) = (\det D(1+T)).$$

Proof. By Lemmas 1.1–1.2 the tower of cokernels is, level by level and compatibly with the transition maps, the tower of presentations $\mathbb{Z}_\ell[G_k]^n \xrightarrow{D(\sigma)} \mathbb{Z}_\ell[G_k]^n \rightarrow C_k \rightarrow 0$ with transitions the coefficient projections. All modules are finitely generated $\mathbb{Z}_\ell$ -modules, hence compact, and the transition maps are continuous surjections on the presenting copies; inverse limits are exact on such systems

(Mittag–Leffler for compact groups), so $\varprojlim C_k = \operatorname{coker}(\varprojlim D(\sigma)) = \Lambda^n / D(1+T)\Lambda^n$. Since $g = \det D(1+T) \neq 0$, the map $D(1+T)$ is injective over the fraction field, so the cokernel is Λ -torsion; it is finitely generated by presentation. For the characteristic ideal, localize at a height-one prime $P \subset \Lambda$: Λ_P is a discrete valuation ring, elementary divisors give $\operatorname{length}_{\Lambda_P}((X_\infty)_P) = v_P(\det D(1+T))$, and taking the product over height-one primes yields $\operatorname{char}_\Lambda(X_\infty) = (\det D(1+T))$ by the structure theory of finitely generated torsion Λ -modules [4]. $\square$

Corollary 2.2 (Exact control). *For every $k \geq 0$, with $\omega_k = (1+T)^{\ell^k} - 1$,*

$$X_\infty / \omega_k X_\infty \cong \operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell \cong (\mathbb{Z} \oplus \operatorname{Jac}(Y_k)) \otimes \mathbb{Z}_\ell,$$

an isomorphism on the nose — not merely up to bounded kernels and cokernels — because $\Lambda / \omega_k \cong \mathbb{Z}_\ell[G_k]$ carries the level- k presentation exactly.

Corollary 2.3 (Invariants, now as a theorem). *The Iwasawa invariants of the tower are $\mu = \mu(g)$ and $\lambda = \lambda(g) - 1$, where $(\mu(g), \lambda(g))$ are read off the Newton polygon of g , the -1 being the trivial-zero and normalization accounting of [II, Rem. 5.4]. In particular the four exact laws of [II, Table 2] are consequences of Theorem 2.1, no longer observations.*

3. THE VERIFICATION BATTERY

tower	levels	(B) block	(S) deck	(P)+(C) norm identities	(N) torsion norm
$(1, 1, 1), \ell = 2$	$k \leq 3$	exact	exact	exact	$\operatorname{surj.}; \ker = 2^3, 2^3, 2^3$
$(1, 0, 0), \ell = 3$	$k \leq 2$	exact	exact	exact	$\operatorname{surj.}; \ker = 3^1, 3^1$

TABLE 1. Machine verification of Lemmas 1.1–1.3: (B) $\Delta_k = D(S_N)$; (S) $R_k \Delta_k = \Delta_k R_k$; (P) $P_k \Delta_{k+1} = \Delta_k P_k$; (C) $P_k R_{k+1} = R_k P_k$; (N) surjectivity of π_* on ℓ -primary torsion with kernel orders exactly $\ell^{\Delta_{\text{ord}}}$. Every check passed as an exact identity; the lemmas are proved above independently, so the table is confirmation, not evidence.

4. RELATION TO THE LITERATURE, STATED PROMINENTLY

The verified bibliographic records of Vallières [1], McGown–Vallières [2] and, above all, Gonet — whose paper is titled precisely *Iwasawa theory of Jacobians of graphs* [3] — make it very likely that Theorem 2.1, or a statement equivalent to it, is present in that literature; the growth law certainly is, as acknowledged since [II, §5]. Under the series’ policy, we have verified those papers’ metadata but have not read their theorems, and we therefore claim priority for nothing in Sections 1–2 beyond what a comparison with their texts will leave standing. What we believe is ours at minimum: the *exactness* of Corollary 2.2 in the form stated, its verification at the resolution of Smith normal forms across [II, Table 2] and Table 1, and the integration of the statement into the trilogy’s K -theoretic framework, where $\operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k})$ is the K_0 -group of the level- k Laplacian channel ([I, Thm. 3.5]), so that Theorem 2.1 reads: *the K -theoretic torsion of the Laplacian channel along an abelian tower is one finitely generated torsion Iwasawa module, with characteristic ideal the voltage pencil determinant.*

5. WHAT REMAINS OPEN

Nothing here touches the two hard problems. The congruence tower is non-normal, its pseudo-Iwasawa law has $\lambda = -1$, and the correct base object — conjecturally a module theory over the Iwahori–Hecke algebra with an intrinsic Euler-characteristic correction ([II, Conj. 3.6(ii-b)], [III,

§2]) — still lacks a definition of characteristic ideal. And the canonicity problem of [III, Rem. 1.10] — a Kirchberg algebra over $C(\Omega)$, functorial in the Hecke data, with $K_0 \cong (\mathbb{Z} \oplus \text{Jac}(Y_k))_k$ — is exactly as open as before; Theorem 2.1 sharpens its target, since the prescribed K -theory is now known to be a single Λ -module, but supplies no construction.

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

Unrefereed preprint. All numbered statements are proved; the machine battery of Table 1 confirms the three lemmas at seven levels across two towers. External inputs: the structure theory of finitely generated torsion Λ -modules and the identification $\varprojlim \mathbb{Z}_\ell[G_k] \cong \Lambda$, both standard, cited to [4], whose bibliographic record is now primary-source verified (only the chapter-level pointer remains a recollection-based reading aid); the graph-Iwasawa literature [1, 2, 3], bibliographic data verified in [II–III], texts unread, priority explicitly ceded in Section 4.

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